Decision-Making Under Uncertainty: Exam Solutions

Contents

Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure may not receive credit. A minimum of 8 points is required to pass each criterion.

Evaluated criteria

- **C1** Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.
- **C2** Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.
- **C3** Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.
- **C4** Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Weekend Market Stall [C1-3 C2-3]

A craft cooperative must decide where to operate its stall for the weekend before knowing how strong customer demand will be. The cooperative can choose either a central stall or a neighbourhood stall, while weekend demand may turn out to be strong or weak.

If the cooperative chooses the central stall, it expects to earn a profit of fourteen thousand pounds when demand is strong and five thousand pounds when demand is weak. If it chooses the neighbourhood stall, it expects to earn a profit of ten thousand pounds when demand is strong and eight thousand pounds when demand is weak.

Questions/Tasks.

- **C1** Characterize the key elements of the decision-making situation. **[3]**
- **C2** Construct the payoff table. **[3]**

C1 - Characterization [3]

Element	Description	
Alternatives	**A1:** Central stall	**A2:** Neighbourhood stall
Statesofnature	**S1:** Strong demand	**S2:** Weak demand
Events	Actual weekend demand after the location decision	
Consequences	Profit in thousand pounds for each alternative–state pair	

C2 - Payoff table [3]

Each stated consequence is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 14, \quad P(A_1, S_2) = 5,$$
$$P(A_2, S_1) = 10, \quad P(A_2, S_2) = 8.$$

Thus, in thousand pounds, the completed payoff table is

Alternative	Strong S_1	Weak S_2
Central stall A_1	14	5
Neighbourhood stall A_2	10	8

2 Urban Delivery Service [C1-5 C2-10]

An urban logistics company provides transportation and delivery services across Bogotá. Before beginning daily operations, the company must decide how to organize the vehicles used to move packages between different parts of the city. It can either hire and operate its own dedicated van fleet or contract a third-party logistics provider to carry out the deliveries. Traffic conditions throughout the city may turn out to be relatively clear or heavily congested.

The payoff from the service depends on the number of deliveries successfully completed during the day, together with contributions and costs associated with both the chosen transportation arrangement and the traffic conditions. All monetary values are measured in thousand USD.

Hiring and operating its own van fleet provides a contribution of 0.7 thousand USD for each completed delivery and involves a fixed cost of 8 thousand USD for the day's operation. Contracting a third-party delivery service instead provides a contribution of 0.5 thousand USD for each completed delivery and involves a fixed cost of 13 thousand USD.

Traffic conditions also affect the financial result of providing the transportation service. When traffic across the city is relatively clear, each completed delivery contributes an additional 0.2 thousand USD, while the operation incurs an additional fixed cost of 4 thousand USD. When the city experiences heavy traffic, each completed delivery contributes an additional 0.1 thousand USD, while the operation incurs a larger additional fixed cost of 6 thousand USD.

The number of deliveries that can be completed during the day also depends on both the transportation arrangement and the traffic conditions. With its own van fleet, the company expects to complete 80 deliveries when traffic is relatively clear and 55 when traffic is heavy. When using a third-party delivery service, it expects to complete 70 deliveries when traffic is relatively clear and 60 when traffic is heavy.

Questions/Tasks.

- **C1** Characterize the key elements of the decision-making situation. **[5]**
- **C2** Construct the payoff table. **[10]**

C1 - Characterization [5]

Element	Description	
Alternatives	**A1:** Own van fleet	**A2:** Third-party logistics provider
Statesofnature	**S1:** Relatively clear traffic	**S2:** Heavy traffic
Events	Actual traffic conditions during daily operations after the transportation arrangement is selected	
Consequences	Profit in thousand USD for each alternative–state pair	

Value type	Context meaning	Symbol	Problem values
Quantity depending on alternative and state	Completed deliveries	$q(A_i, S_j)$	$q(A_1, S_1) = 80,\ q(A_1, S_2) = 55$ $q(A_2, S_1) = 70,\ q(A_2, S_2) = 60$
Alternative-based variable revenue	Contribution per completed delivery associated with the transportation arrangement	$r^A(A_i)$	$r^A(A_1) = 0.7,\ r^A(A_2) = 0.5$
State-based variable revenue	Additional contribution per completed delivery associated with traffic conditions	$r^S(S_j)$	$r^S(S_1) = 0.2,\ r^S(S_2) = 0.1$
Alternative-based fixed cost	Daily fixed operating cost associated with the transportation arrangement	$F^{C,A}(A_i)$	$F^{C,A}(A_1) = 8,\ F^{C,A}(A_2) = 13$
State-based fixed cost	Additional daily fixed cost associated with traffic conditions	$F^{C,S}(S_j)$	$F^{C,S}(S_1) = 4,\ F^{C,S}(S_2) = 6$

Monetary rates and fixed amounts are in thousand USD; the final payoff is $P(A_i, S_j)$.

C2 - Payoff calculations and table **[10]**

For every cell,

$$P(A_i, S_j) = q(A_i, S_j)[r^A(A_i) + r^S(S_j)] \\ - F^{C,A}(A_i) - F^{C,S}(S_j).$$

Own fleet, clear traffic: combined contribution $= 0.7 + 0.2 = 0.9$ per delivery; variable contribution $= 80(0.9) = 72$; combined fixed cost $= 8 + 4 = 12$. Therefore

$$P(A_1, S_1) = 72 - 12 = 60 \text{ thousand USD.}$$

Own fleet, heavy traffic: combined contribution $= 0.7 + 0.1 = 0.8$; variable contribution $= 55(0.8) = 44$; combined fixed cost $= 8 + 6 = 14$. Thus

$$P(A_1, S_2) = 44 - 14 = 30 \text{ thousand USD.}$$

Third party, clear traffic: combined contribution $= 0.5 + 0.2 = 0.7$; variable contribution $= 70(0.7) = 49$; combined fixed cost $= 13 + 4 = 17$. Thus

$$P(A_2, S_1) = 49 - 17 = 32 \text{ thousand USD.}$$

Third party, heavy traffic: combined contribution $= 0.5 + 0.1 = 0.6$; variable contribution $= 60(0.6) = 36$; combined fixed cost $= 13 + 6 = 19$. Thus

$$P(A_2, S_2) = 36 - 19 = 17 \text{ thousand USD.}$$

The completed payoff table, in thousand USD, is

Alternative	Clear S_1	Heavy S_2
Own fleet A_1	60	30
Third party A_2	32	17

3 School Lunch Service [C1-3 C3-3]

A school enterprise must decide which lunch menu to offer before it knows how strong student demand will be. The enterprise can prepare either a hot-lunch menu or a sandwich menu, while demand may turn out to be high or low.

If the enterprise chooses the hot-lunch menu and demand is high, it expects total revenue of 24 thousand USD and total costs of 10 thousand USD. If demand is low, the same menu is expected to generate total revenue of 15 thousand USD and total costs of 9 thousand USD.

If the enterprise chooses the sandwich menu and demand is high, it expects total revenue of 21 thousand USD and total costs of 12 thousand USD. If demand is low, the sandwich menu is expected to generate total revenue of 18 thousand USD and total costs of 11 thousand USD.

Questions/Tasks.

- **C1** Characterize the key elements of the decision-making situation. **[3]**
- **C3** Calculate the profit, then construct the payoff table. **[3]**

C1 - Characterization [3]

Element	Description	
Alternatives	**A1:** Hot-lunch menu	**A2:** Sandwich menu
Statesofnature	**S1:** High student demand	**S2:** Low student demand
Events	Actual student demand after the lunch menu is selected	
Consequences	Profit in thousand USD for each alternative–state pair	

Value type	Context meaning	Symbol	Problem values
Direct total revenue	Total lunch-menu revenue for each menu–demand pair	$R(A_i, S_j)$	$R(A_1, S_1) = 24,\ R(A_1, S_2) = 15$ $R(A_2, S_1) = 21,\ R(A_2, S_2) = 18$
Direct total cost	Total lunch-menu cost for each menu–demand pair	$C(A_i, S_j)$	$C(A_1, S_1) = 10,\ C(A_1, S_2) = 9$ $C(A_2, S_1) = 12,\ C(A_2, S_2) = 11$

All problem values are in thousand USD, and $P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$.

C3 - Payoff calculations and table [3]

Use

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

For the hot-lunch menu under high demand, $R(A_1, S_1) = 24$ and $C(A_1, S_1) = 10$, so

$$P(A_1, S_1) = 24 - 10 = 14 \text{ thousand USD.}$$

Under low demand, $R(A_1, S_2) = 15$ and $C(A_1, S_2) = 9$, so

$$P(A_1, S_2) = 15 - 9 = 6 \text{ thousand USD.}$$

For the sandwich menu under high demand, $R(A_2, S_1) = 21$ and $C(A_2, S_1) = 12$, so

$$P(A_2, S_1) = 21 - 12 = 9 \text{ thousand USD.}$$

Under low demand, $R(A_2, S_2) = 18$ and $C(A_2, S_2) = 11$, so

$$P(A_2, S_2) = 18 - 11 = 7 \text{ thousand USD.}$$

The completed payoff table, in thousand USD, is

Alternative	High S_1	Low S_2
Hot lunch A_1	14	6
Sandwich A_2	9	7

4 Cold-Chain Distribution [C1-5 C3-10]

A food distributor must decide which refrigerated fleet to operate before it knows what demand and fuel conditions will be. The company can choose either a compact refrigerated fleet or a large refrigerated fleet. After this decision is made, market and fuel conditions may turn out to be favourable or adverse.

The number of delivery batches that can be handled depends on both the fleet selected and the conditions that occur. With the compact refrigerated fleet, the distributor expects to handle 800 delivery batches under favourable conditions and 300 under adverse conditions. With the large refrigerated fleet, it expects to handle 1,100 delivery batches under favourable conditions and 400 under adverse conditions.

Revenue from each 100 delivery batches depends partly on the fleet selected. The compact refrigerated fleet generates $12,000 per 100 batches, while the large refrigerated fleet generates $16,000 per 100 batches. Revenue is also affected by the conditions that occur. Favourable conditions add $3,000 of revenue per 100 batches, while adverse conditions add $1,000.

The fleet choice also creates variable costs for each 100 delivery batches. The compact refrigerated fleet has a variable cost of $5,000 per 100 batches, while the large refrigerated fleet has a variable cost of $7,000 per 100 batches. Conditions create an additional variable cost as well. Favourable conditions add $1,000 of cost per 100 batches, while adverse conditions add $4,000.

In addition to these variable amounts, the fleet selected generates fixed revenues and fixed costs. Choosing the compact refrigerated fleet provides $16,000 in fixed revenue and creates $20,000 in fixed costs. Choosing the large refrigerated fleet provides $25,000 in fixed revenue and creates $35,000 in fixed costs.

The conditions that occur also generate fixed financial effects. Favourable conditions provide $12,000 in additional fixed revenue and create $6,000 in additional fixed costs. Adverse conditions provide $4,000 in additional fixed revenue and create $14,000 in additional fixed costs.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[5]**

C3 Calculate the profit, then construct the payoff table. **[10]**

C1 - Characterization **[5]**

Element	Description	
Alternatives	**A1:** Compact refrigerated fleet	**A2:** Large refrigerated fleet
Statesofnature	**S1:** Favourable market and fuel conditions	**S2:** Adverse market and fuel conditions
Events	Actual market demand and fuel conditions after the refrigerated fleet is selected	
Consequences	Profit in thousand USD for each alternative–state pair	

Value type	Context meaning	Symbol	Problem values
Quantity depending on alternative and state	Hundreds of delivery batches handled (batches divided by 100)	$q(A_i, S_j)$	$q(A_1, S_1) = 800/100 = 8,\ q(A_1, S_2) = 300/100 = 3$ $q(A_2, S_1) = 1{,}100/100 = 11,\ q(A_2, S_2) = 400/100 = 4$
Alternative-based variable revenue	Revenue per 100 batches associated with fleet size	$r^A(A_i)$	$r^A(A_1) = 12,\ r^A(A_2) = 16$
State-based variable revenue	Additional revenue per 100 batches associated with market and fuel conditions	$r^S(S_j)$	$r^S(S_1) = 3,\ r^S(S_2) = 1$
Alternative-based fixed revenue	Fixed revenue associated with fleet size	$F^{R,A}(A_i)$	$F^{R,A}(A_1) = 16,\ F^{R,A}(A_2) = 25$
State-based fixed revenue	Additional fixed revenue associated with market and fuel conditions	$F^{R,S}(S_j)$	$F^{R,S}(S_1) = 12,\ F^{R,S}(S_2) = 4$
Alternative-based variable cost	Cost per 100 batches associated with fleet size	$c^A(A_i)$	$c^A(A_1) = 5,\ c^A(A_2) = 7$
State-based variable cost	Additional cost per 100 batches associated with market and fuel conditions	$c^S(S_j)$	$c^S(S_1) = 1,\ c^S(S_2) = 4$
Alternative-based fixed cost	Fixed cost associated with fleet size	$F^{C,A}(A_i)$	$F^{C,A}(A_1) = 20,\ F^{C,A}(A_2) = 35$
State-based fixed cost	Additional fixed cost associated with market and fuel conditions	$F^{C,S}(S_j)$	$F^{C,S}(S_1) = 6,\ F^{C,S}(S_2) = 14$

All monetary rates and fixed amounts are in thousand USD; variable rates apply per 100 batches.

C3 - Payoff calculations and table **[10]**

For each pair,

$$\begin{aligned} R &= q[r^A(A_i) + r^S(S_j)] \\ &\quad + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C &= q[c^A(A_i) + c^S(S_j)] \\ &\quad + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P &= R - C. \end{aligned}$$

All following monetary results are in thousand USD.

Compact / favourable (A_1, S_1). Here $q = 800/100 = 8$. The combined revenue rate is $12 + 3 = 15$, so variable revenue is $8(15) = 120$, and

$$R(A_1, S_1) = 120 + 16 + 12 = 148.$$

The combined cost rate is $5 + 1 = 6$, so variable cost is $8(6) = 48$, and

$$C(A_1, S_1) = 48 + 20 + 6 = 74.$$

Therefore $P(A_1, S_1) = 148 - 74 = 74$ thousand USD.

Compact / adverse (A_1, S_2). Here $q = 300/100 = 3$. The combined revenue rate is $12 + 1 = 13$, variable revenue is $3(13) = 39$, and

$$R(A_1, S_2) = 39 + 16 + 4 = 59.$$

The combined cost rate is $5 + 4 = 9$, variable cost is $3(9) = 27$, and

$$C(A_1, S_2) = 27 + 20 + 14 = 61.$$

Therefore $P(A_1, S_2) = 59 - 61 = -2$ thousand USD, a loss of 2 thousand USD.

Large / favourable (A_2, S_1). Here $q = 1100/100 = 11$. The combined revenue rate is $16 + 3 = 19$, variable revenue is $11(19) = 209$, and

$$R(A_2, S_1) = 209 + 25 + 12 = 246.$$

The combined cost rate is $7 + 1 = 8$, variable cost is $11(8) = 88$, and

$$C(A_2, S_1) = 88 + 35 + 6 = 129.$$

Therefore $P(A_2, S_1) = 246 - 129 = 117$ thousand USD.

Large / adverse (A_2, S_2). Here $q = 400/100 = 4$. The combined revenue rate is $16 + 1 = 17$, variable revenue is $4(17) = 68$, and

$$R(A_2, S_2) = 68 + 25 + 4 = 97.$$

The combined cost rate is $7 + 4 = 11$, variable cost is $4(11) = 44$, and

$$C(A_2, S_2) = 44 + 35 + 14 = 93.$$

Therefore $P(A_2, S_2) = 97 - 93 = 4$ thousand USD.

The completed payoff table, in thousand USD, is

Alternative	Favourable	Adverse
Compact A_1	74	-2
Large A_2	117	4

5 Mobile Produce Shop [C4-2]

A produce seller must locate a mobile shop at the business district (A_1) or the residential district (A_2). Demand may be high (S_1) or low (S_2). Profits, in thousand USD, are:

Alternative	S_1	S_2
Business district	48	12
Residential district	35	24

Questions/Tasks.

C4.1 Apply Maximax and state the selected location. **[1]**

C4.2 Apply Maximin and state the selected location. **[1]**

C4.1 - Maximax [1]

This optimistic criterion selects $\max_i[\max_j P(A_i, S_j)]$. The row maxima are

$$\begin{aligned} \max_j P(A_1, S_j) &= \max\{48, 12\} = 48 \\ \max_j P(A_2, S_j) &= \max\{35, 24\} = 35 \end{aligned} \quad \longrightarrow \quad \max\{48, 35\} = 48$$

Therefore Maximax selects the $\boldsymbol{A_1}$ **: Business district**.

C4.2 - Maximin [1]

This conservative criterion selects $\max_i[\min_j P(A_i, S_j)]$. The row minima are

$$\begin{aligned} \min_j P(A_1, S_j) &= \min\{48, 12\} = 12 \\ \min_j P(A_2, S_j) &= \min\{35, 24\} = 24 \end{aligned} \quad \longrightarrow \quad \max\{12, 24\} = 24$$

Therefore Maximin selects the $\boldsymbol{A_2}$ **: Residential district**.

6 Warehouse Technology [C4-8]

A wholesaler must choose one of four warehouse systems before market and supply conditions are known. The possible states are strong demand (S_1), steady demand (S_2), and weak demand with delays (S_3). The profits, in thousand USD, are:

Alternative	S_1	S_2	S_3
Automated system	96	52	18
Modular system	78	61	42
Robotic system	88	48	30
Flexible system	72	64	50

Questions/Tasks.

C4.1 Apply Maximax and state the selected location. **[4]**

C4.2 Apply Maximin and state the selected location. **[4]**

4.1 - Maximax **[4]**

This optimistic criterion selects $\max_i[\max_j P(A_i, S_j)]$. The row maxima are

$$\begin{aligned}
\max_j P(A_1, S_j) &= \max\{96, 52, 18\} = 96 \\
\max_j P(A_2, S_j) &= \max\{78, 61, 42\} = 78 \\
\max_j P(A_3, S_j) &= \max\{88, 48, 30\} = 88 \\
\max_j P(A_4, S_j) &= \max\{72, 64, 50\} = 72
\end{aligned} \quad \longrightarrow \quad \max\{96, 78, 88, 72\} = 96$$

Therefore Maximax selects the $\boldsymbol{A_1}$ **: Automated system**.

4.2 - Maximin **[4]**

This conservative criterion selects $\max_i[\min_j P(A_i, S_j)]$. The row minima are

$$\begin{aligned}
\min_j P(A_1, S_j) &= \min\{96, 52, 18\} = 18 \\
\min_j P(A_2, S_j) &= \min\{78, 61, 42\} = 42 \\
\min_j P(A_3, S_j) &= \min\{88, 48, 30\} = 30 \\
\min_j P(A_4, S_j) &= \min\{72, 64, 50\} = 50
\end{aligned} \quad \longrightarrow \quad \max\{18, 42, 30, 50\} = 50$$

Therefore Maximin selects the $\boldsymbol{A_4}$ **: flexible system**.